

RESEARCH ARTICLE SUMMARY

MESOSCOPIC PHYSICS

Time-domain braiding of anyons

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INTRODUCTION: Unlike fermions and bosons, anyons are quasiparticles that retain a robust memory of particle exchanges through a braiding phase factor, a property called fractional statistics. This memory is at the heart of topological quantum computing, where quantum information can be processed by braiding anyons. It also provides anyons with distinctive dynamical properties that are so far unexplored. Anyon fractional statistics have recently been evidenced by several experiments that all focused on the stationary regime. As such, they did not provide any information on anyon dynamics. However, when manipulating anyons at the single-particle level, it becomes important to understand how braiding affects the timescales of anyon transfer in a circuit.

RATIONALE: In a fractional quantum Hall (FQH) fluid, braiding plays a prominent role when studying the scattering of an anyon emitted toward a narrow constriction called a quantum point contact (QPC). In this case, the dominant mechanism for particle transfer across the constriction is not the direct tunneling of the incoming excitation, but rather a braiding process between the incoming excitation and an anyon–anti-anyon pair created at the QPC. The creation of the pair, triggered by the incoming excitation, can be understood as a tunneling event across the QPC where an anyon is created on one side, leaving an anyon hole on the other. As a consequence of the robust memory of braiding, these tunneling events may occur long after the incoming anyon has exited the QPC. By contrast, if the incoming particle is an electron with trivial braiding, tunneling occurs at the same time as the electron crosses the QPC. In this work, we used a triggered source of anyon pulses (of temporal width of a few tens of picoseconds) to investigate the dynamics of anyon tunneling at a QPC in a FQH fluid at filling factor $\nu = 1/3$. In the experiment, we used Hong-Ou-Mandel

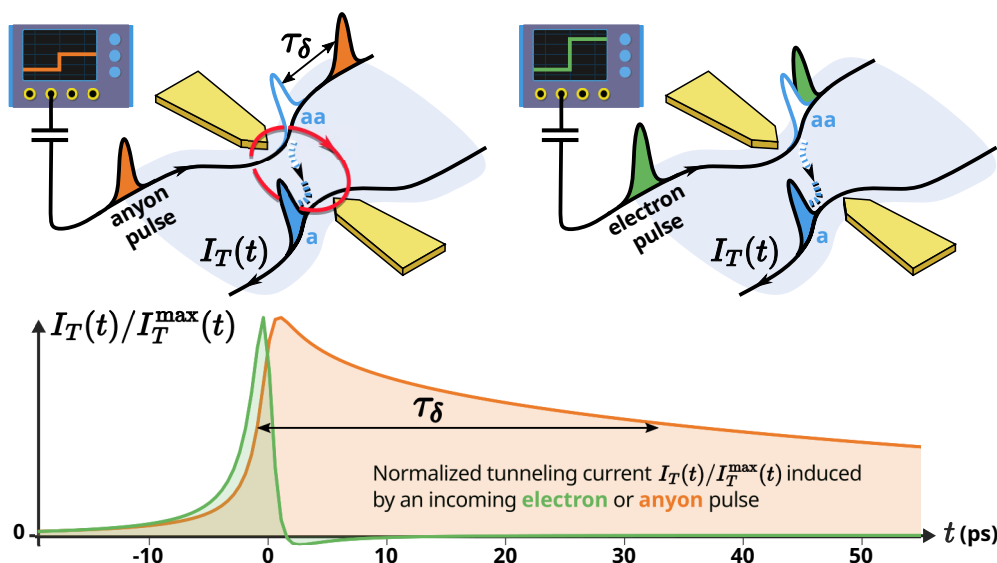
interferometry, where two mutually time-shifted anyon pulses are emitted toward the QPC. By measuring the correlations of the electrical current fluctuations at the QPC output as a function of the time delay between pulses, we directly probed the characteristic anyon tunneling timescale τ_δ .

RESULTS: We observed that anyon braiding increases the characteristic timescale τ_δ for anyon tunneling. Our findings demonstrate that τ_δ is set by the temporal decay of the anyon correlation function and can be understood as the characteristic time at which the anyon memory is erased. It grows when the temperature and the scaling dimension δ (the parameter characterizing the dynamics at the edge of a FQH fluid) decrease. Our study allowed us to extract the value of $\delta = 0.66 \pm 0.08$ by measuring the evolution of the tunneling dynamics with temperature. It differed from the expected value for a Laughlin state, $\delta = 1/3$, indicating that δ is nonuniversal. By varying the number of anyons contained in each pulse, we observed that braiding effects are suppressed for pulses containing three anyons, confirming the quantized value of $\theta = 2\pi/3$ of the braiding phase at filling factor $1/3$.

CONCLUSION: Our work provides an experimental signature of the role of braiding in anyon dynamics, highlighting the distinctive properties of anyons compared with those of fermions or bosons. It introduces time-domain measurements for characterizing the properties of anyons—their braiding phase and their scaling dimension—therefore setting the stage for further exploration of more complex FQH states such as those hosting non-Abelian anyons. □

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Tunneling dynamics of electrons and anyons. (Top left) A generator emits an anyon pulse (orange) propagating toward the QPC (gold). Anyon braiding leads to the tunneling of an anyon–anti-anyon pair (a-aa; blue) long after the anyon pulse has exited the QPC. (Top right) For electron emission (green), tunneling occurs immediately after the electron crosses the QPC. Consequently, the tunneling current $I_T(t)$ decays rapidly for electrons and slowly for anyons.



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Time-domain braiding of anyons

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B. Grémaud², T. Jonckheere², T. Martin², J. Rech², A. Cavanna³,
U. Gennser³, Y. Jin³, G. Ménard¹, G. Fève^{1*}

Anyons are quasiparticles that keep a robust memory of particle exchanges via a braiding phase factor. When an anyon excitation is emitted toward a quantum point contact (QPC) in a fractional quantum Hall (FQH) fluid, this memory translates into tunneling events that may occur long after the anyon excitation has exited the QPC. In this work, we used triggered anyon pulses incident on a QPC in a filling factor $\nu = 1/3$ FQH fluid to investigate anyon tunneling in the time domain. We observed that braiding increases the tunneling timescale, which is set by the temperature and the anyon scaling dimension that characterizes the edge-state dynamics. Our experiment introduces time-domain measurements for characterizing the braiding phase and scaling dimension of anyons.

Two-dimensional systems can host anyons (1–4), topological excitations that are neither fermions nor bosons and obey intermediate, fractional statistics. In the simplest case of Abelian anyons, fractional statistics implies that moving one anyon around another [a braiding operation (5)] results in the accumulation of a nontrivial phase factor $e^{\theta} \neq 1$. For non-Abelian anyons (6, 7), the order of exchanging the particles also plays a role, forming the basis of topological quantum computing. Fractional quantum Hall conductors (FQHCs) (8) have been suggested as systems of interest for hosting anyons as the elementary excitations of their insulating bulk (9, 10). However, probing the bulk is experimentally challenging, and most characterizations of FQHCs have been performed by using transport measurements along their conducting edge channels. By weakly coupling two counterpropagating edge channels with a quantum point contact (QPC), one can randomly transfer anyon excitations from one edge to the other. As they are transferred through the insulating FQH fluid at the narrow gap of the QPC, these excitations inherit the topological properties of the bulk with a quantized fractional charge e^* and braiding phase θ .

More than 20 years ago, single QPCs were used for the determination of the fractional charge e^* (11–13). More complex experiments relying on single-particle (14–16) and two-particle (17–21) interferometry and using sample geometries comprising several QPCs have been suggested to probe anyon fractional statistics. Following these proposals, fractional statistics of anyons were experimentally demonstrated in 2020 at the filling factor $\nu = 1/3$ in collider (22) and Fabry-Perot geometries (23), which was followed by several experiments (24–29) confirming the previous results and extending them to other filling factors. Apart from e^* and θ , anyon dynamics at the edge of a FQHC are also characterized by a third parameter, the scaling dimension δ (30, 31), which sets the characteristic time of the temporal correlations of edge anyonic excitations. Unlike the fractional charge and statistics that are imposed by the topological properties of the bulk, δ only reflects the properties of the edge. Therefore, it is not necessarily quantized and may depend on local parameters such as the

interaction strength and the smoothness of the edge (32). Experimental attempts at extracting δ have relied on the measurement of the non-linear current (I)–voltage (V) characteristics resulting from anyon tunneling at a QPC (33–35). Despite qualitative similarities with theory predictions within the Luttinger model (30), no quantitative agreement was observed (36).

Almost all previous experiments investigating anyon properties have focused on the dc regime using stationary anyon currents. As such, they do not offer the possibility to trigger the emission of anyons in the circuit in order to probe their dynamics. Controlling anyon emission at the single-particle level is crucial for probing anyon braiding in the time domain, performing on-demand braiding operations, and accessing the characteristic timescales of anyon transfer in a circuit. Furthermore, by bringing additional information with respect to their dc counterpart, time-resolved experiments provide a way to disentangle the respective roles of the different properties of anyons, such as their fractional charge, statistics, and scaling dimension. However, emitting single anyons on demand is challenging. Deterministic electron emission has been realized in the integer quantum Hall regime by using a driven quantum dot (37), but these methods cannot be applied to the anyon case. There is a mutual incompatibility between the confinement in a quantum dot and the emission of anyons, as the necessary barrier favors the tunneling of electrons over anyons. There have been suggestions to circumvent this problem by using antidots (38), but a simpler route was recently proposed. It consists of applying fast time-dependent drives to a one-dimensional chiral edge channel (39–41) via a metallic gate or an ohmic contact. Owing to their gapless and linear low-energy spectrum, chiral edge channels host collective bosonic excitations (42), which consist in a continuous variation of the local charge density that propagates along the edge in a chiral manner. By driving a one-dimensional edge channel with a short-time voltage pulse, one can thus generate a propagating current pulse $I(x, t)$ carrying

a total charge $q = \int_{-\infty}^{+\infty} I(x, t) dt$ that varies continuously with the am-

plitude (and width) of the applied drive. This implies that, unlike a tunnel process, the number of anyons carried per pulse $N = q/e^*$ is not quantized to a discrete number but can be controlled with the drive amplitude, offering a knob for tuning anyon braiding in time-resolved experiments.

In this work, we investigated the properties of anyon tunneling at a QPC in a $\nu = 1/3$ FQHC: their braiding phase θ characterizing their exchange properties and their scaling dimension δ characterizing their dynamical properties. To achieve this, we generated short-time anyon current pulses characterized by their temporal width W (here fixed to ≈ 100 ps) and by the number of anyons N carried by each pulse. We investigated anyon braiding by studying the partitioning of the generated anyon pulses at a QPC. When anyons are impinging on a QPC, the dominant mechanism for particle transfer is not the direct tunneling of the incoming excitations, but rather a braiding process between the incoming excitations and particle-hole excitations created at the QPC (41, 43–45). Anyon tunneling is then governed by the mutual braiding phase $\theta \times N$ (46) between the generated anyon pulses and the topological anyons tunneling at the QPC. In contrast to previous experiments, the mutual braiding phase can be varied by tuning the dimensionless charge N carried by each anyon pulse. Additionally, the triggering of anyon emission allowed us to investigate anyon braiding in the time domain and to characterize its effect on the dynamics of anyon tunneling at the QPC. We observed that for a nontrivial braiding ($\theta \times N < 2\pi$), anyons keep a memory of braiding processes occurring at the QPC, and tunneling may happen long after the emitted anyon pulse has exited the QPC. The characteristic timescale for anyon tunneling is then set by the temporal decay of the anyon correlation function over a time $\tau_\delta = \hbar/(\pi k_B T e \delta)$, parameterized by the Planck and Boltzmann constants $\hbar$ and k_B , the electronic

¹Laboratoire de Physique de l'Ecole normale supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université Paris Cité, Paris, France. ²Aix Marseille Univ, Université de Toulon, CNRS, CPT, Marseille, France. ³Centre de Nanosciences et de Nanotechnologies (C2N), CNRS, Université Paris-Saclay, Palaiseau, France. *Corresponding author. Email: gwendal.feve@ens.fr

temperature T_{el} , and the scaling dimension δ . It can be understood as the characteristic time at which the anyon memory is erased at the edge of the FQH fluid. It implies that the effect of braiding on anyon tunneling dynamics is more important at low electronic temperature and for small values of δ . By measuring the dependence of τ_δ on the temperature T_{el} , we extracted $\delta = 0.66 \pm 0.08$ for a wide range of QPC parameters. It differs from the naïve expectation $\delta = \nu = 1/3$ for a Laughlin state (30), showing that δ is indeed nonuniversal. By contrast, when braiding is trivial ($\theta \times N = 2\pi$), we observed that the characteristic tunneling timescale is set by the temporal width of the generated pulses, as one would naïvely expect. For pulses of narrow width $W \approx 100$ ps, this is observed as a reduction by a factor of two of the characteristic tunneling time when braiding becomes trivial. Finally, by measuring that the suppression of braiding effects occurs for pulses containing $N = 3$ anyons, we confirmed the quantized value of the braiding phase of topological anyons, $\theta = 2\pi/3$ at filling factor $1/3$. By giving immediate access to the scaling dimension δ and to the anyon braiding phase θ , our experiment illustrates the relevance of triggered anyon emission, allowing us to explore the consequences of fractional statistics. These results will be instrumental for the characterization of topological anyon excitations in more complex FQHCs, where the fractional charge and the statistics are not characterized by the same quantized number.

Braiding at a QPC and anyon pulses

Anyon tunneling between counterpropagating edge channels can be induced by using a QPC. In the limit of weak backscattering, only two tunneling processes, represented in Fig. 1A in the equilibrium situation, need to be considered. The left panel represents the tunneling at time t' of an anyon (in dark blue) from channel 1 to channel 2, leaving a hole (in light blue) in channel 1. This tunnel process can be represented by the quantum state $|t'\rangle_+$, where the sign “+” stands for the direction of tunneling (from channel 1 to channel 2). The associated tunneling rate $\Gamma_+(t)$ results from the time-domain interference (41, 43–45) (see also fig. S2) between tunneling events occurring at two different times t and t' and can be written as the following quantum overlap:

$$\Gamma_+(t) \propto \Re \left[\int dt' \langle t|t' \rangle_+ \right]. \text{ Together with the}$$

forward tunneling $|t'\rangle_+$, one needs also to consider backward tunneling events $|t'\rangle_-$ describing the transfer of an anyon at time t' from channel 2 to channel 1 (Fig. 1A, right panel), which occurs with a rate

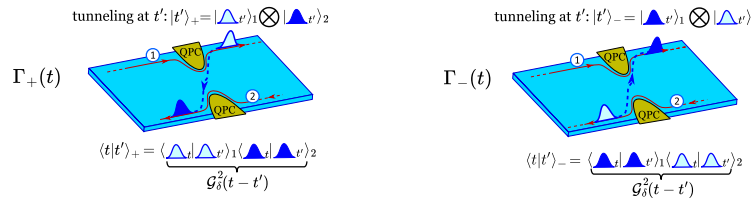
$$\Gamma_-(t) \propto \Re \left[\int dt' \langle t|t' \rangle_- \right]. \text{ At equilibrium,}$$

where no anyon sources are connected at the input of the QPC, the two tunneling rates are equal, $\Gamma_+(t) = \Gamma_-(t)$, and describe random transfers of anyons from one edge to the other. The magnitude of the equilibrium tunneling rates is then related to the equilibrium correlation function of the FQH fluid: $\langle t|t' \rangle_+ = \langle t|t' \rangle_- = \mathcal{G}_\delta^2(t-t')$, which decays on long times with the characteristic timescale τ_δ . The circumstances are

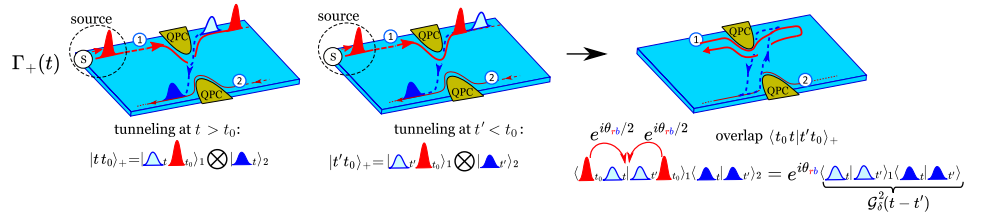
completely different when considering the nonequilibrium case where an anyon source is connected at input 1 of the QPC. Figure 1B represents the forward tunneling process occurring with rate $\Gamma_+(t)$ in the simple case where a single anyon excitation (in red) is emitted by the source and reaches the QPC at time t_0 . In this case, $\Gamma_+(t)$ is governed by the braiding processes occurring between the red anyon excitation emitted by the source and the anyon quasi-hole excitation created in channel 1 by the anyon tunneling process. This braiding mechanism is illustrated in Fig. 1B, where the left panel (respectively, center panel) represents the quantum state $|t t_0\rangle_+$ (respectively, $|t' t_0\rangle_+$) describing forward anyon tunneling (from channel 1 to channel 2) at time t (respectively, t') in the presence of a red anyon reaching the QPC at time t_0 . For $t' \leq t_0 \leq t$, the red anyon crosses the QPC before the tunneling event at time t (left panel) but after the tunneling event at time t' (center panel). Anyon braiding thus emerges from the time-domain interferometry (41, 43–45) (see also fig. S2) between the two tunneling processes $|t t_0\rangle_+$ and $|t' t_0\rangle_+$. As illustrated in the right panel of Fig. 1B, the overlap $\langle t_0 t | t' t_0 \rangle_+$ differs from the equilibrium value by a relative phase given by the mutual braiding phase between the blue anyon quasi-hole and the red anyons having crossed the QPC in the time window $t - t'$.

This braiding phase thus depends on the nature of the blue and red quasiparticles. The blue anyons tunneling at the QPC are the topological excitations whose properties are imposed by the bulk topological order,

A Equilibrium fluctuations at a QPC



B Time-domain braiding : anyon source in channel 1



C Topological anyon tunneling at input QPC: $\theta_{rb} = \theta$

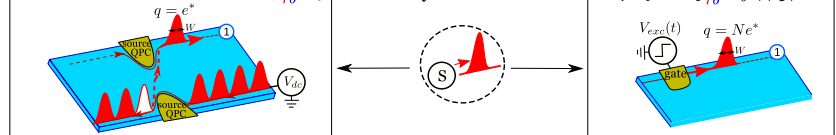


Fig. 1. Anyon braiding at a QPC. (A) Anyon tunneling at a QPC, equilibrium case. Shown on the left is the forward tunneling process described by the state $|t'\rangle_+$, where an anyon (in dark blue) tunnels from channel 1 to channel 2 at time t' , leaving a hole (in light blue) in channel 1. The forward tunneling rate $\Gamma_+(t)$ results from the time-domain interference between the different tunneling processes $|t\rangle_+$ and $|t'\rangle_+$ occurring at different times and can be computed from the overlap: $\Gamma_+(t) = \Re \left[\int dt' \langle t|t' \rangle_+ \right]$. Shown on the right is the backward tunneling process described by the state $|t'\rangle_-$, where an anyon tunnels from channel 2 to channel 1 at time t' . At equilibrium, $\Gamma_+(t) = \Gamma_-(t)$ and the tunneling current $I_T(t) = e^*[\Gamma_+(t) - \Gamma_-(t)]$ vanishes. (B) Anyon braiding when an anyon source is connected at input 1. In the presence of an anyon excitation (in red) incoming on the QPC at time t_0 , the processes $|t t_0\rangle_+$ (left) and $|t' t_0\rangle_+$ (center) represent the tunneling of the blue anyon after (for $t \geq t_0$) and before (for $t' \leq t_0$) the red anyon has crossed the QPC. The overlap $\langle t_0 t | t' t_0 \rangle_+$ is sketched in the right panel, where a reversed time propagation is used to represent the bra $\langle t_0 t | \langle t' t_0 \rangle_+$, which differs from the overlap at equilibrium $\langle t|t' \rangle_+$ by the braiding phase factor $e^{i\theta_{tb}}$, where θ_{tb} is the mutual braiding phase between the red and blue anyons [see (46) for a more detailed explanation]. (C) Two different sources of anyons may be used. In anyon collider experiments (left), a source QPC biased by a dc voltage generates a dilute and stationary beam of anyons from anyon tunneling. θ_{rb} thus takes the quantized value θ imposed by the bulk topological order. In this work, we investigated the triggered emission of an anyon pulse with charge $q = Ne^*$ from a voltage-driven metallic top gate (right). The mutual braiding then equals $\theta_{rb} = \theta \times N$, where N is the number of anyons carried per pulse.

whereas the nature of the red anyons depends on the anyon source. In anyon collider experiments (22, 24–26), the red anyons are also randomly generated by a tunneling process at a source QPC driven by a dc voltage (Fig. 1C) and are thus also topological anyons. The mutual braiding phase is thus quantized to the value θ , which has proved decisive for the study of anyon fractional statistics in collider experiments. We investigated here the emission of current pulses via a metallic gate (Fig. 1C, right panel), each pulse carrying a charge $q = Ne^*$. As a result, the mutual braiding phase between the blue and red anyons is $\theta \times N$ (46). When N is an integer, this phase can be straightforwardly interpreted as stemming from the fusion of N anyons (7). Interestingly, the gapless spectrum of edge channels allows us to generalize this expression to noninteger values of N , so that the braiding phase can be continuously tuned in experiments.

In the case of the backward tunneling rate $\Gamma_{-}(t)$, discussed in more detail in (46), the anyon tunneling process occurs in the opposite direction (from channel 2 to channel 1). The red anyon braids with an anyon quasiparticle instead of an anyon quasihole, resulting in an opposite braiding phase compared with that of the forward tunneling case. The time-dependent tunneling rates $\Gamma_{+}(t)$ and $\Gamma_{-}(t)$, as well as the tunneling current through the QPC $I_T(t) = e^*[\Gamma_{+}(t) - \Gamma_{-}(t)]$, can then be directly expressed as a function of the braiding phase:

$$\Gamma_{\pm}(t) = C \Re \left[\int_{-\infty}^t dt' e^{\pm i\theta \times N(t',t)} \mathcal{G}_{\delta}^2(t-t') \right] \quad (1)$$

$$I_T(t) = -2e^*C \int_{-\infty}^t dt' \sin[\theta \times N(t',t)] \Im [\mathcal{G}_{\delta}^2(t-t')] \quad (2)$$

where C is a constant. As discussed above, the accumulated braiding phase $\theta \times N(t', t)$ in a time window $t - t'$ is set by the number of emitted anyons $N(t', t)$ crossing the QPC between times t' and t :

$$N(t', t) = \int_{t'}^t I(\tau) / e^* d\tau, \text{ where } I(t) \text{ is the electrical current generated}$$

by the anyon source connected at input 1. As seen from Eq. 1, the edge correlation function $\mathcal{G}_{\delta}(t-t')$ sets the characteristic time window $t - t'$ allowing for braiding mechanisms to take place. This can be understood from the braiding picture of Fig. 1B: The time uncertainty $t - t'$ for the creation of the particle-hole excitation at the QPC is typically set by the equilibrium correlation time at the edge of the FQHC τ_{δ} . The characteristic timescale on which braiding processes may take place thus increases when decreasing T_{el} or decreasing the scaling dimension δ .

Tunneling dynamics and Hong-Ou-Mandel experiment

Equation 1 shows that the tunneling dynamics are governed by two timescales. The first one is the characteristic time for the evolution of the time-dependent braiding phase $\theta \times N(t', t)$. It is set by the temporal width of the anyon pulses W . The second one is the characteristic timescale τ_{δ} of the edge correlation function $\mathcal{G}_{\delta}$. To simplify the respective roles of these two timescales, we consider the case of very narrow pulses, $W \ll \tau_{\delta}$. Two different situations may occur, depending on the number of anyons in each pulse N . Let us first consider the simpler case where braiding is trivial, $\theta \times N = 2\pi$. Figure 2A represents the evolution of the time-dependent braiding phase $\theta \times N(t, t')$ when varying t at fixed $t' < t_0$. At time t_0 where the pulse crosses the QPC, the braiding phase jumps from zero to the value $\theta \times N = 2\pi$ in the short timescale W of the pulse width. As a result, the factor $\sin[\theta \times N(t', t)]$ in the expression of $I_T(t)$ in Eq. 2 jumps from zero back to zero within the time W (Fig. 2A). This shows that when braiding is trivial, the characteristic timescale for tunneling is set by the short temporal width of the emitted pulses. This contrasts with the situation where braiding mechanisms are nontrivial: $\theta \times N < 2\pi$. In this case, the braiding phase jumps from zero to the value $\theta \times N < 2\pi$ when the anyon pulse crosses the QPC (see Fig. 2A in the case $N = 1$), and $\sin[\theta \times N(t', t)]$

evolves from zero to the constant, nonzero value $\sin(\theta \times N)$ during the short time W . The current then falls back to zero on the longer timescale τ_{δ} associated with the decay of the correlation function $\mathcal{G}_{\delta}(t-t')$. The characteristic timescale of the tunneling current is thus set by the long time τ_{δ} on which braiding processes may take place. This is a consequence of the fundamental property of anyons: They keep a memory of the exchange processes taking place in tunneling at a QPC. This memory, which is encoded as a braiding induced phase shift, is erased on the characteristic timescale τ_{δ} for the decay of correlations at the edge of the FQH fluid. This implies that by using triggered sources of anyons, it is possible to provide direct evidence of anyon braiding in the time domain by measuring the slow decay of $I_T(t)$ on the timescale τ_{δ} . Figure 2B represents the numerically computed time evolution of $I_T(t)$ from Eq. 2 using $\theta = 2\pi/3$ (the expected braiding phase at $\nu = 1/3$) and for a pulse containing $N = 1$ anyon ($\theta \times N = 2\pi/3$). The decay of $I_T(t)$ is longer than the short time W . It also directly depends on the value of δ , as illustrated by the difference between $\delta = 1/3$ (brown curve) and $\delta = 2/3$ (orange curve). This contrasts with the case of pulses containing $N = 3$ anyons (or equivalently one electron at filling factor $\nu = 1/3$) represented by the green curve. In this case, braiding is trivial ($\theta \times N = 2\pi$ for $\theta = 2\pi/3$) and the tunneling current decays on the short timescale W .

Time-domain measurements of the tunneling current using fast-triggered anyon sources can thus probe both the anyon braiding phase and the scaling dimension. However, measurements of the time-dependent electrical current with a time resolution on the order of 10 ps are extremely difficult. As proposed in (40), by emitting a second anyon pulse of opposite charge $-N$ at the input of the QPC after a tunable time delay D (with a 15-ps resolution), we can recover the dynamics of $\Gamma_{\pm}(t)$ from dc measurements of the time-averaged tunneling rates as a function of D , $\bar{\Gamma}_{\pm}(D)$. The role of the second anyon pulse is to cancel the braiding phase accumulated by the passing of the first anyon pulse (see Fig. 2C). Indeed, an anyon and an anti-anyon fused together have zero braiding phase with a topological anyon tunneling at the QPC. However, this cancellation occurs only in the limit of short time delays (Fig. 2C, top panel), where D is much shorter than the characteristic time for $\Gamma_{\pm}(t)$ to decay down to zero: τ_{δ} for $\theta \times N < 2\pi$ and W for $\theta \times N = 2\pi$. In this limit, the tunneling rates take their equilibrium value. $D \gg \tau_{\delta}$ (when braiding is nontrivial) or $D \gg W$ (when braiding is trivial) corresponds to the opposite limit of long time delays (Fig. 2C, bottom panel). In this case, the two anyon pulses give rise to two independent tunneling events governed by the braiding of each pulse with the topological anyons tunneling at the QPC. This corresponds to the maximum deviation of $\bar{\Gamma}_{\pm}(D)$ from their equilibrium value. This experiment shares strong similarities with Hong-Ou-Mandel (HOM) experiments performed in optics (47) or electronics (48) where two photons or two electrons are emitted toward a beam splitter with a tunable time delay between the two particles. In the bosonic (47) and fermionic (48) cases where braiding is trivial, HOM experiments were used to measure the temporal width W of the emitted pulses. In the anyonic case, a new timescale τ_{δ} appears in the dynamics of anyon transfer owing to their nontrivial braiding properties.

The two experimentally accessible quantities related to $\bar{\Gamma}_{\pm}(D)$ are the tunneling current $I_T(D) = e^*(\bar{\Gamma}_{+} - \bar{\Gamma}_{-})$ and its fluctuations $S_T(D) = 2(e^*)^2(\bar{\Gamma}_{+} + \bar{\Gamma}_{-})$. Measuring both quantities simultaneously provides a stringent test of the role of braiding on the tunneling dynamics as I_T and S_T probe complementary aspects of the edge correlations: I_T probes the imaginary part of $\mathcal{G}_{\delta}(t-t')$, whereas S_T probes the real part. However, the particle-hole symmetry of the emitted anyon pulses implies that $I_T(D)$ vanishes for all time delays (as $\bar{\Gamma}_{+} = \bar{\Gamma}_{-}$). It is thus necessary to induce a small asymmetry between the two tunneling rates by applying a small dc voltage V at input 2 of the QPC to restore a nonzero tunneling current. The small additional stream of tunneling anyons generated by the dc voltage is then used as a noninvasive probe of the tunneling rates $\bar{\Gamma}_{+/-}$ and of their dependence on

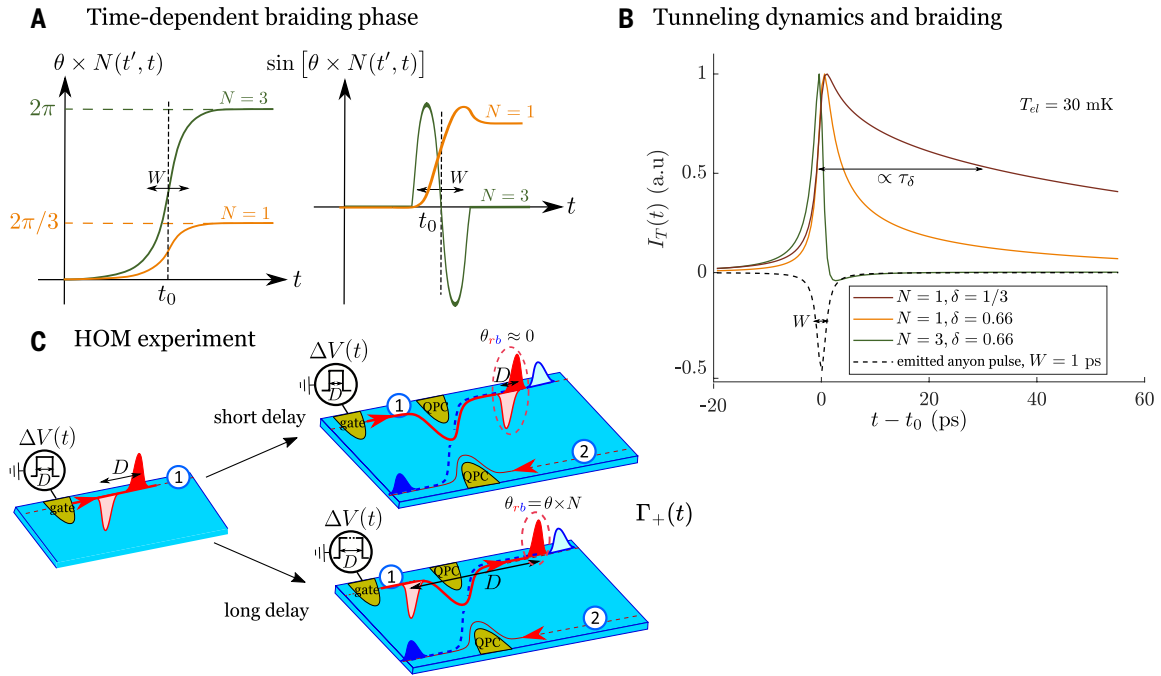


Fig. 2. Time-dependent braiding phase and principle of the HOM experiment. (A) Shown on the left is a sketch of the time-dependent braiding phase $\theta \times N(t', t)$, where $N(t', t)$ is the number of anyons crossing the QPC between times t' and t . It is plotted as a function of t with fixed $t' < t_0$. We compare pulses containing $N = 1$ (orange trace) and $N = 3$ (green trace) anyons, with $\theta = 2\pi/3$. In both cases, the braiding phase jumps from zero to $\theta \times N$ at the time $t = t_0$ where the anyon pulse crosses the QPC and on the timescale W (width of the anyon pulse). Shown on the right is a sketch of $\sin[\theta \times N(t', t)]$ as a function of t . For $N = 1$, it jumps to the constant value $\sin(2\pi/3)$ and stays constant afterward, illustrating the memory left in the system for nontrivial braiding. By contrast, $\sin[\theta \times N(t', t)]$ varies from zero back to $\sin(2\pi) = 0$ over time W for $N = 3$ (trivial braiding). (B) Simulations of $I_T(t)$ using Eq. 2. For $N = 3$ (trivial braiding, green line), the current decays on the short timescale W , the temporal width of the emitted pulses (dashed black line). For $N = 1$, the decay takes place over the longer time τ_δ owing to the nontrivial braiding and depends on the value of δ ($1/3$ for the brown line and 0.66 for the orange line). a.u., arbitrary units. (C) Principle of the HOM experiment. Two anyon pulses of opposite signs and separated by a tunable time delay D are emitted at input 1 of the QPC by applying a time-dependent square voltage to the gate. For short time delays D (top right), the two pulses can be considered as a single entity with zero braiding phase, where the tunneling topological anyons and $\Gamma_\pm(t)$ are equal to their equilibrium value. For long time delays D (bottom right), the two pulses give rise to two independent tunneling processes with opposite values of the braiding phase for the particle-like and hole-like pulses. This is the regime where $\Gamma_\pm(t)$ deviate the most from their equilibrium value. The transition between the regimes of short and long delays occurs on the characteristic timescale for $\Gamma_\pm(t)$ to decay down to zero: W when braiding is trivial ($\theta \times N = 2\pi$), and τ_δ when braiding is nontrivial ($\theta \times N \neq 2\pi$).

the delay D through the measurement of the differential conductance $G_T(D) = I_T(D, V)/V$. In this work, we investigated both the excess noise $\Delta S_T(D)$ and the excess conductance $\Delta G_T(D)$ compared with equilibrium (anyon sources switched off, or equivalently $D = 0$). By using Eq. 1 to compute both quantities, their dependence on the time-delay D allowed us to probe two complementary aspects of the tunneling dynamics:

$$\Delta S_T(D) = 4C(e^*)^2 f \int_0^{1/f} d\tau \int_0^{+\infty} d\tau' \{\cos[\theta \times \Delta N(D)] - 1\} \Re[C_\delta^2(\tau)] \quad (3)$$

$$\Delta G_T(D) = -2C \frac{(e^*)^2}{\hbar} f \int_0^{1/f} d\tau \int_0^{+\infty} d\tau' \{\cos[\theta \times \Delta N(D)] - 1\} \Im[\tau C_\delta^2(\tau)] \quad (4)$$

where $\theta \times \Delta N(D) = \theta \times \int_{t-\tau/2}^{t+\tau/2} \frac{I(t') - I(t' - D)}{e^*} dt'$ is the time-dependent braiding phase resulting from the two anyon pulses, and f is the pulse emission frequency.

Sample

The sample, shown in Fig. 3, is a two-dimensional electron gas (GaAs/AlGaAs) of density $n_s = 1.1 \times 10^{15} \text{ m}^{-2}$ and mobility $\mu = 1.4 \times 10^6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, set at filling factor $\nu = 1/3$ by applying a perpendicular

magnetic field $B = 13.6 \text{ T}$ (fig. S3). Applying a step voltage to the metallic gate G1 generates a current pulse propagating along edge channel 1 (49). Its charge can be tuned by the amplitude V_{exc} of the excitation drive, and its shape can be approximated by a square (46) of width imposed by the transit time below the gate, $W \approx 100 \text{ ps}$ (with a gate length of approximately $1.7 \mu\text{m}$ and a charge velocity of $2 \times 10^4 \text{ m s}^{-1}$ that we deduced from the measured pulse width). The second anyon pulse is generated by applying a second step voltage of opposite sign after a time delay D on gate G1. The emission of both pulses is then repeated at a frequency $f = 0.65 \text{ GHz}$, with a period $1/f \approx 1.5 \text{ ns}$ chosen to be much larger than both W and τ_δ . The two pulses then propagate toward the central QPC (cQPC) for the investigation of the tunneling dynamics. The current noise S_4 is measured on output 4 of cQPC. Note that ΔS_4 differs from ΔS_T by the additional contribution of the thermal noise transmitted through the QPC: $\Delta S_4 = \Delta S_T - 4k_B T_{\text{el}} \Delta G_T$. This difference becomes relevant if G_T varies when anyons are emitted, which is the case in the experiment. G_T is measured by applying a small ac voltage at low frequency (1 MHz) on gate G2 and measuring the tunneling current through the QPC at output 3. We defined the two normalized HOM experimental signals as $s(D) = \frac{\Delta S_4(D)}{\Delta S_4[D \approx 1/(2f)]}$ and $g(D) = \frac{\Delta G_T(D)}{\Delta G_T[D \approx 1/(2f)]}$. By construction, s and g vary from zero in the short delay limit ($D = 0$) to 1 in the long delay limit ($D \approx 1/(2f)$), allowing us to focus on the width of $s(D)$ and $g(D)$, which contains the information on the two characteristic timescales, τ_δ and W .

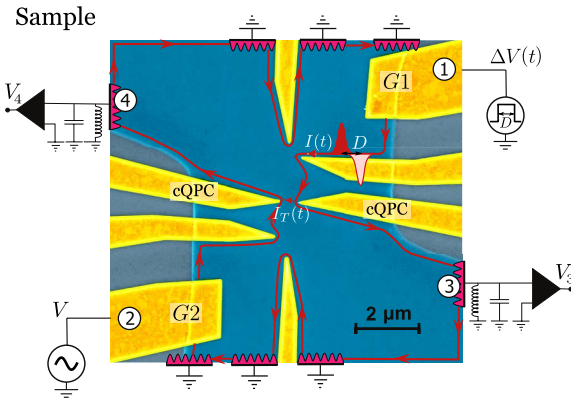


Fig. 3. Sample. The electron gas is represented in blue, the edge channels in red, and the metallic gates in gold. By applying a square voltage at a frequency $f = 0.65$ GHz to gate G1, an anyon pulse (in red) followed by a pulse of opposite charge (in pink with a red edge) are emitted at input 1 of the cQPC. The time delay D can be tuned by varying the width of the square excitation. The differential conductance G_T is measured by applying a small ac voltage (at a frequency of 1 MHz) to gate G2 and by measuring the resulting tunneling current $I_T(t)$ on output 3 of the cQPC. The noise S_4 is measured at output 4 of the cQPC using homemade cryogenic amplifiers.

Braiding of anyons in the time domain

We started by calibrating the charge carried by each pulse by measuring the non-normalized output noise $\Delta S_T[D \approx 1/(2f)]$ when the two pulses of charge Ne^* are maximally separated [$D \approx 1/(2f)$]. Figure 4, A and B, represents $\Delta S_T[D \approx 1/(2f)]$ and its derivative $\frac{d\Delta S_T[D \approx 1/(2f)]}{dV_{exc}}$ as a function of the voltage drive amplitude V_{exc} . $N = q/e^*$ is linearly related to V_{exc} by an unknown lever arm α , $V_{exc} = \alpha N$. The value of α is adjusted for the best agreement between the data and the theory prediction of Eq. 3 [see (46) for details on the comparison with theory]. As can be seen in Fig. 4A, good agreement can be obtained up to $N \approx 3$ using $\alpha = 180$ mV. In particular, using this value reproduces the maximum value of $\frac{d\Delta S_T[D \approx 1/(2f)]}{dV_{exc}}$ observed for $N \approx 1.5$ in Fig. 4B. The uncertainty on the position of the maximum is used for the estimation of the error bar on the calibration, $\alpha = 180 \pm 30$ mV. Above $N \approx 3$, discrepancies

between data and model can be seen in Fig. 4, A and B, which can probably be attributed to heating effects caused by the ac excitation.

We now turn to the measurements of the HOM traces s and g in the two different situations where braiding is nontrivial ($\theta \times N < 2\pi$) and where braiding is trivial ($\theta \times N = 2\pi$). For the first case, we chose $N = 1$, where a single anyon excitation is carried by each pulse, which reproduces the conditions of collider experiments. For the second case, we chose $N = 3$, which is expected to cancel braiding effects for $\theta = 2\pi/3$ at $\nu = 1/3$. This can also be understood as corresponding to a single electron per pulse, given that electrons have a 2π braiding phase with anyon excitations at $\nu = 1/3$. Figure 5, A to C, presents the measurements of $s(D)$ (left panel) and $g(D)$ (right panel) for three different temperatures, $T_{el} = 30, 40$, and 60 mK, respectively. The green points correspond to $N = 1$ anyon per pulse, and the orange points correspond to $N = 3$ anyons. All traces exhibit the common features of the HOM effect, with a dip in $s(D)$ and $g(D)$ located around $D = 0$. At $T_{el} = 30$ mK, a clear difference can be seen between the $N = 1$ and $N = 3$ cases. The HOM dip in the $N = 3$ case is much narrower compared with that in the $N = 1$ case, showing that the characteristic timescale for tunneling is shorter for $N = 3$ (where braiding is trivial) compared with $N = 1$ (where braiding is nontrivial). For a quantitative measurement of the width difference, we extracted the width at half height $D_{1/2}$ of $s(D)$ for $N = 1$ and $N = 3$ from a Lorentzian fit. For $N = 3$, the narrow HOM dip $D_{1/2}(N = 3) = 40 \pm 1.5$ ps reflects the temporal width W of the current pulses propagating toward the QPC. For $N = 1$, the dip increases to $D_{1/2}(N = 1) = 79 \pm 4$ ps, providing a direct signature, in the time domain, of the role of braiding effects on anyon tunneling. We observed that, at the lowest temperature reached in our experiment, braiding effects increased the characteristic tunneling timescale by a factor of two. This effect can be seen in both s and g , with different measured shapes of the HOM dips $s(D)$ and $g(D)$. These differences reflect the fact that s and g probe different aspects of edge dynamics: $\Re[\mathcal{G}_s^2(\tau)]$ for s and $\Im[\mathcal{G}_g^2(\tau)]$ for g (see Eqs. 3 and 4). Increasing the temperature suppresses the differences between $N = 1$ and $N = 3$, which become very faint at $T_{el} = 60$ mK. This decay is expected as $\tau_\delta = \hbar/(\pi k_B T_{el} \delta)$ decreases when increasing the temperature, so that the widening of $I_T(t)$ caused by braiding effects vanishes. For a quantitative test of the braiding scenario, all HOM traces $s(D)$ and $g(D)$ for $N = 1$ and $N = 3$ as well as for different temperatures are compared with numerical evaluations of Eqs. 3 and 4 [dashed lines; see (46) for details

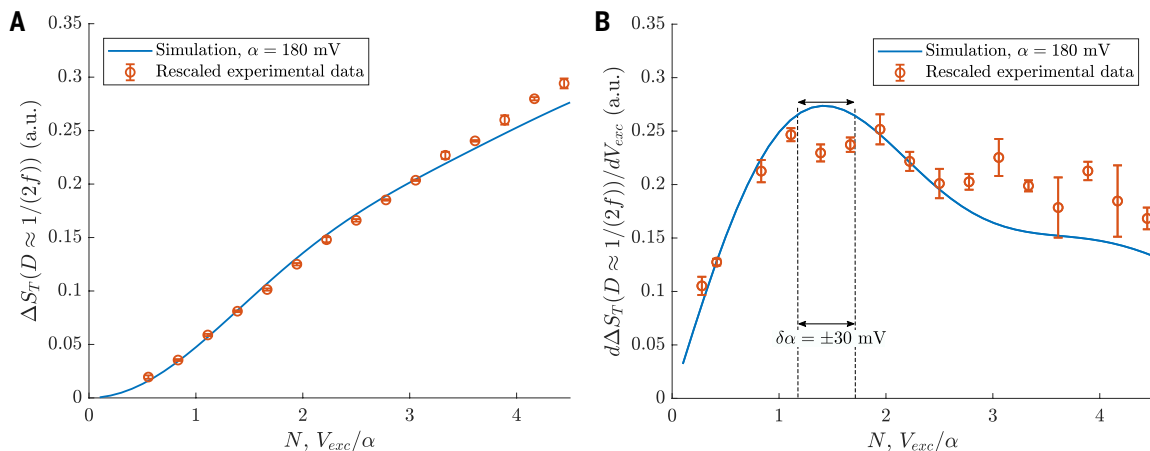


Fig. 4. Calibration of the dimensionless charge N . (A) Measurement of the output noise ΔS_T for a perfect separation between the pulses, $D = 1/(2f)$, as a function of the amplitude $V_{exc}/\alpha = N$ of the step voltage generated by our arbitrary wave generator at room temperature. The blue trace is the numerical calculation $\Delta S_T(N)$ evaluated using Eq. 3 with the following parameters: $T_{el} = 45$ mK, $W = 123$ ps, and $\delta = 0.66$. The lever arm $\alpha = 180 \pm 30$ mV is adjusted for the best agreement with the data for $N \leq 3$. (B) $d\Delta S_T/dV_{exc}$ as a function of V_{exc}/α . The blue trace is the numerical calculation using Eq. 3 with the same parameters as above. The maximum of $d\Delta S_T/dV_{exc}$ at $N \approx 1.5$ is used for the calibration of α . Error bars indicate the standard error of the mean.

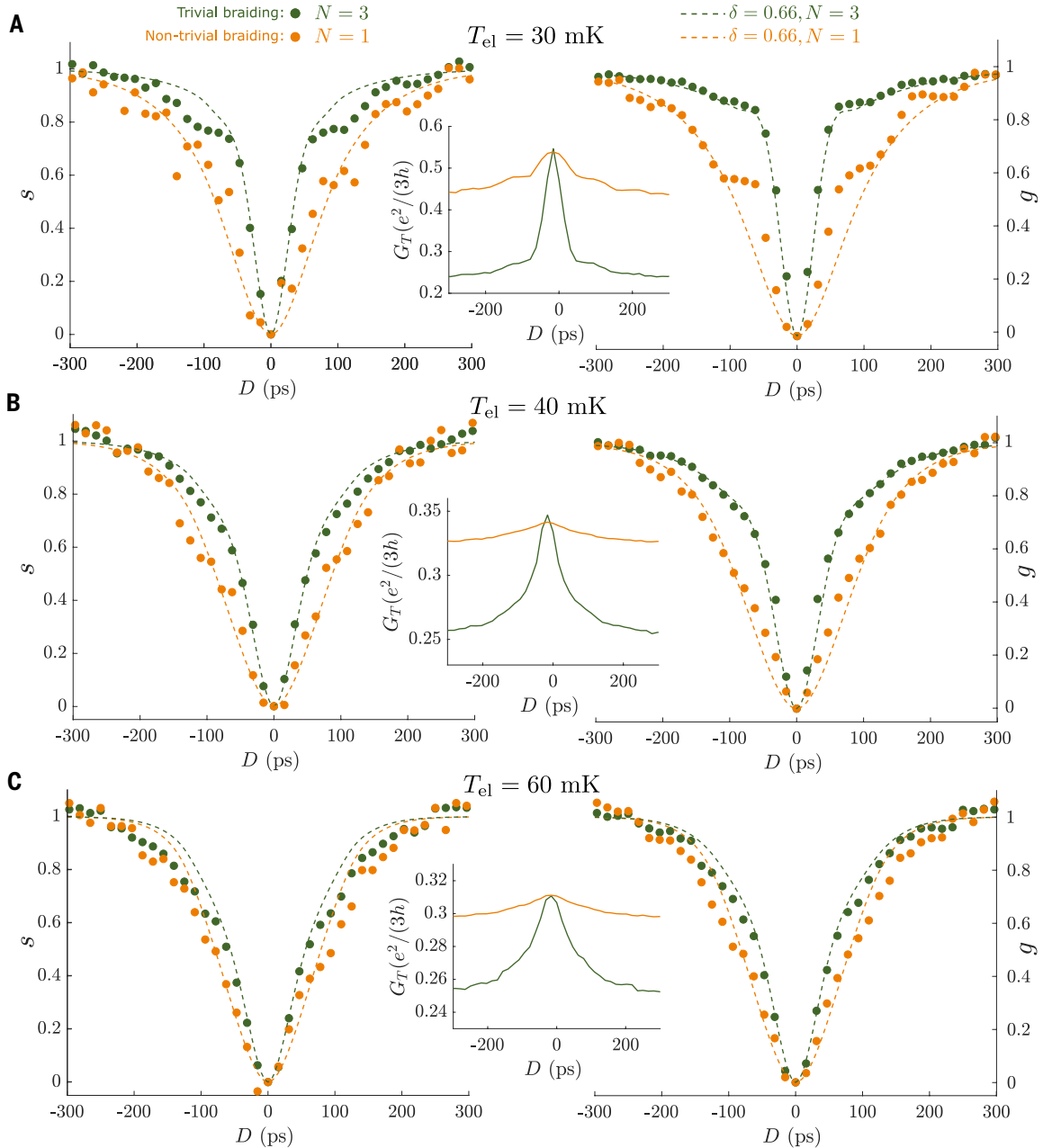


Fig. 5. Braiding of anyons in the time domain. (A and B) Measurements of $s(D)$ (left panels) and $g(D)$ (right panels) for pulses of charge $N=1$ (orange points) and $N=3$ (green points) at different temperatures. A clear widening of the HOM dip can be observed between $N=1$ and $N=3$ at the lowest temperature, $T_{el} = 30$ mK (A). It demonstrates that nontrivial braiding increases the characteristic tunneling timescale. In other words, when braiding is present, tunneling events keep occurring long after the anyon pulse has exited the QPC. The width difference between $N=1$ and $N=3$ progressively decreases when increasing the temperature to $T_{el} = 40$ mK (B) and 60 mK (C). It reflects that the characteristic timescale of braiding effects τ_δ varies as $1/T_{el}$. Measurements are well reproduced by numerical simulations (green and orange dashed lines) of Eqs. 3 and 4 with the following parameters: $W = 0.065/f = 100$ ps and $\delta = 0.66$ for (A), and $W = 0.083/f = 128$ ps and $\delta = 0.66$ for (B) and (C). The insets at the center represent the dimensionless differential conductance $G_T/(e^2/3h)$ as a function of D .

of the comparison with theory]. We took $\mathcal{G}_\delta(\tau)$ from the Luttinger model (30), $\mathcal{G}_\delta(\tau) \propto \frac{1}{\{\sinh[\frac{1}{2} \ln k_B T_{el}(\tau_0 + i\tau)/\hbar]\}^\delta}$, where τ_0 is a short time cut-

off, and we took $\delta = 0.66$. The only remaining unknown parameter was the width W of the emitted current pulses. It was adjusted for the best agreement of $g(D)$ for $N=3$, giving $W = 100$ ps at $T_{el} = 30$ mK and $W = 128$ ps at $T_{el} = 40$ and 60 mK. The discrepancy between the two values can be explained by the fact that the measurements at $T_{el} =$

40 and 60 mK were taken in the same experimental run, whereas the measurements at $T_{el} = 30$ mK were obtained in a subsequent run after the removal of bias tees at cryogenic temperatures to generate sharper excitation pulses. The overall agreement is very good. In particular, the differences between the dip widths for s and g , as well as the suppression of braiding effects when increasing the temperature, are well captured. This demonstrates that for $N=1$, braiding mechanisms slow down the tunneling dynamics compared with the $N=3$ case, where

braiding is trivial. Some discrepancies can be observed, which are most pronounced at $T_{\text{el}} = 30$ mK. They are probably related to some additional structure of the emitted current pulses that can easily occur when sending ultrashort pulses in a quantum conductor.

For an accurate estimation of the scaling dimension δ , we measured the evolution of the width at half $D_{1/2}(N = 1)$ of the HOM dip of $s(D)$ at $N = 1$ as a function of temperature. To eliminate the contribution of the width of the emitted current pulses W , we plot in Fig. 6A $\Delta D_{1/2} = D_{1/2}(N = 1) - D_{1/2}(N = 3)$ as a function of the temperature T_{el} . As discussed above, $\Delta D_{1/2}$ decreases when increasing the temperature and the decay rate can be directly compared with the predictions of Eq. 3 for different values of δ ($\delta = 0.66$ is the blue line, $\delta = 0.58$ the blue dashed line, and $\delta = 0.74$ the blue dashed-dotted line). The best agreement with theory is obtained for $\delta = 0.66 \pm 0.08$.

To probe the value of the braiding phase θ at $\nu = 1/3$ from our time-domain experiment, we measured the continuous evolution of the width of the HOM dip for s and g as a function of the dimensionless charge N carried by the anyon pulses. Because the widths of the HOM dips for s and g are different (see Fig. 5), we defined the normalized width $\Delta \tilde{D}_{1/2}(N) = \frac{D_{1/2}(N) - D_{1/2}(N=3)}{D_{1/2}(N=1) - D_{1/2}(N=3)}$ which takes almost identical values for s and g (the small differences predicted by theory are much smaller than the experimental resolution). Figure 6B displays the measurements of the normalized excess width for the HOM traces of both the noise $s(D)$ (red points) and the differential conductance $g(D)$ (violet and yellow points). For $N \leq 1$, the normalized width is close to 1 and then decays smoothly toward zero as N reaches 3. This behavior is well reproduced by the model (blue dashed line) using $\theta = 2\pi/3$. This shows that braiding effects can be suppressed by generating pulses containing $N = 3$ anyons, implying $\theta \times 3 = 2\pi$ and thus confirming the value $\theta = 2\pi/3$ of the braiding phase at $\nu = 1/3$. This demonstrates the value of time-resolved measurements for probing the properties of anyons. A better accuracy for the measurement of the braiding phase could be obtained by improving the calibration of the number of anyons per pulse N and by generating anyon pulses with shorter temporal width W . Indeed, when varying N , the sharpness of the transition between

the regime dominated by braiding [$\Delta \tilde{D}_{1/2}(N) = 1$] and the regime where braiding is trivial [$\Delta \tilde{D}_{1/2}(N) = 0$] is directly controlled by the temporal width of the anyon pulses (fig. S6): The sharper the pulses are, the more abruptly the cancellation of braiding effects happens.

We finally investigated how the HOM experimental signals vary when tuning the QPC gate voltage that sets the QPC backscattering probability. For this study, we focused on the measurement of g , which is faster to carry out (an additional measurement of s for $G_T < 0.1 \frac{e^2}{h}$ is shown in fig. S4). Figure 7A presents all our measurements of $G_T(D)$ for $N = 1$ and $N = 3$ and for different settings of the QPC backscattering probability P_0 defined at zero time delay: $P_0 = G_T(D = 0)/(e^2/3h)$. Interestingly, the measurements of $G_T(D)$ for $P_0 > 0.1$ agree qualitatively with the predictions of the Luttinger model for the nonlinear evolution of the differential conductance G_T (decrease of G_T when increasing the time delay D). By contrast, the measurements of $G_T(D)$ for $P_0 = 0.06$ exhibit an anti-Luttinger dependence (G_T increases when increasing D). This behavior, which is systematically observed in the very weak backscattering regime in I - V characteristic measurements, would suggest an inconsistent value of $\delta > 1$. By contrast, all our measurements of $g(D)$ for $N = 1$ and $N = 3$ plotted in Fig. 7B collapse onto the same curves, with the same increase of the width of the HOM dips for all values of P_0 . This shows that by probing the scaling dimension directly in the time domain, our measurement of $\delta \approx 0.66$ is robust to the variations of the QPC backscattering probability P_0 .

Discussion and outlook

We investigated the scaling dimension and braiding phase of topological anyons tunneling at a QPC at filling factor $\nu = 1/3$ by studying the partitioning of short pulses carrying a tunable number of anyons N . By emitting two mutually time-delayed anyon pulses of opposite charge in a HOM setup, we measured the characteristic timescale for anyon tunneling at the QPC. We compared two situations: $N = 1$ where braiding is nontrivial (with a mutual braiding phase $\theta \times 1 = 2\pi/3$ at $\nu = 1/3$), and $N = 3$ where braiding effects are expected to be suppressed at $\nu = 1/3$ ($\theta \times 3 = 2\pi$). At the lowest temperature $T_{\text{el}} = 30$ mK, we observed a widening by a factor 2 of the HOM dip between $N = 3$ and $N = 1$. This shows that nontrivial braiding increases the characteristic tunneling time, being governed at the edge of the FQHC by the temporal correlations $\tau_\delta = \hbar/(\pi k_B T_{\text{el}} \delta)$. Our time-domain measurements thus provide a direct experimental signature of the role of braiding in anyon dynamics, illustrating that anyons have specific physical properties, distinct from those of fermions or bosons. From the temperature dependence of the width difference $\Delta D_{1/2}(T_{\text{el}})$ of the HOM dips between $N = 1$ and $N = 3$, we extracted the scaling dimension $\delta = 0.66 \pm 0.08$. By measuring the evolution of the normalized width $\Delta \tilde{D}_{1/2}(N)$ with the number of anyons per pulse N , we characterized the suppression of braiding effects on the dynamics of anyon tunneling when $\theta \times N$ approaches 2π . In particular, our measurements show that braiding effects become trivial for $N = 3$, confirming previous measurements of $\theta = 2\pi/3$ at $\nu = 1/3$ (22–26).

Our measured value of $\delta = 0.66 \pm 0.08$ contrasts with the expectation of $\delta = \nu = 1/3$ for a Laughlin FQHC, showing that δ is nonuniversal (32). Our experiment

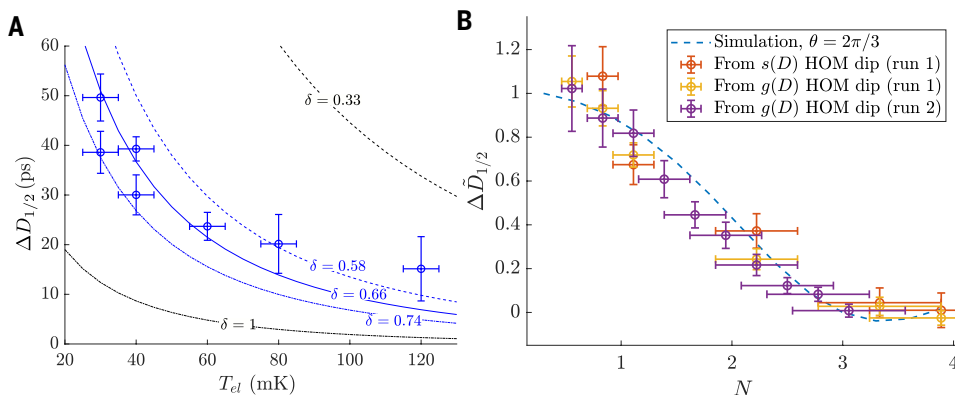


Fig. 6. Extraction of the scaling dimension δ and braiding phase θ . (A) Excess width $\Delta D_{1/2} = D_{1/2}(N = 1) - D_{1/2}(N = 3)$ of the HOM dips of $s(D)$ as a function of the temperature T_{el} . $\Delta D_{1/2}$ decreases with increasing temperature. This shows that the memory of braiding effects is erased on a timescale set by the temporal correlations at the edge of the quantum fluid that are inversely proportional to the temperature and to the anyon scaling dimension δ . The lines are numerical simulations with $W = 100$ ps, $\delta = 0.66$ (blue line), $\delta = 0.58$ (blue dashed line), $\delta = 0.74$ (blue dashed-dotted line), $\delta = 1/3$ (black dashed line), and $\delta = 1$ (black dashed-dotted line). The best agreement is obtained for $\delta = 0.66 \pm 0.08$. (B) Normalized excess width $\Delta \tilde{D}_{1/2} = \frac{D_{1/2}(N) - D_{1/2}(N=3)}{D_{1/2}(N=1) - D_{1/2}(N=3)}$ as a function of the anyon charge N . The red points are measurements of s at $T_{\text{el}} = 40$ mK (and $W = 128$ ps), the yellow points are measurements of g at $T_{\text{el}} = 40$ mK (and $W = 128$ ps), the violet points are measurements of g at $T_{\text{el}} = 30$ mK (and $W = 100$ ps). The width is predicted to be minimum when braiding effects are suppressed ($\theta \times N = 2\pi$), allowing for a determination of the braiding phase θ . Our measurements are in good agreement with the prediction for $\theta = 2\pi/3$ (blue dashed lines). Parameters for the numerical simulation are $T_{\text{el}} = 30$ mK, $W = 100$ ps, and $\delta = 0.66$. Error bars indicate the standard error of the mean.

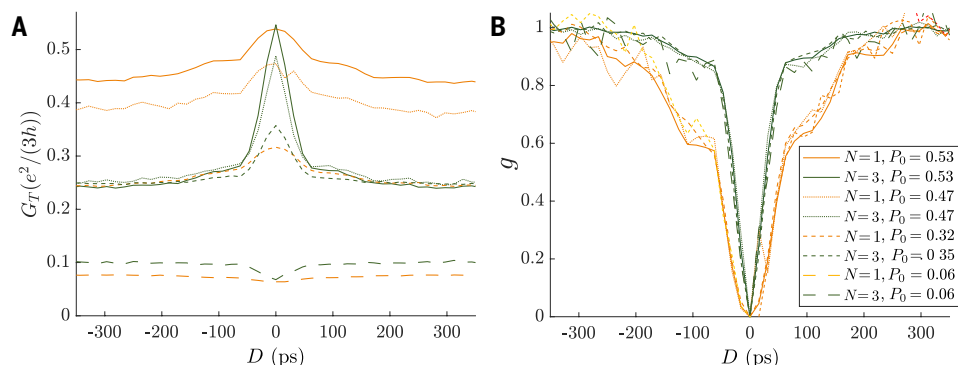


Fig. 7. Evolution of the scaling dimension with the QPC backscattering probability. (A) Measurements of $G_T(D)$ for $N=1$ and $N=3$ for different values of the backscattering probability ($T_{el}=30$ mK). (B) Normalized dimensionless conductance $g(D)$ for $N=1$ and $N=3$ corresponding to the measurements of $G_T(D)$ in (A). Despite the measured differences between the different traces of $G_T(D)$, all traces of $g(D)$ for $N=1$ and $N=3$ fall on the same curve. In particular, the difference of the width of the HOM dips between $N=1$ and $N=3$ is the same for the different traces corresponding to different settings of the QPC backscattering probability. It shows that the scaling dimension δ is rather insensitive to the specific value of the QPC backscattering probability.

raises the question of the control and engineering of the dynamical properties of the edge channels set by δ : Can it be modified by changing the nature of the edge, the quantum point contact geometry, or the nature of the conductor (GaAs versus graphene)? Regarding the characterization of anyon properties in a FQHC, our experiment allowed us to disentangle the effect of the scaling dimension δ , which controls the characteristic timescale when braiding is nontrivial, from that of the braiding phase θ , which controls the nontrivial or trivial nature of braiding mechanisms in tunneling experiments. This will thus be instrumental for characterizing anyon braiding in more complex Abelian FQH states where the fractional charge e^* and the braiding phase θ are given by different quantized numbers. For example, at $\nu=2/5$, anyons have a fractional charge $e^*=e/5$ and a braiding phase $\theta=3/5 \times 2\pi$, meaning that pulses containing $N=5/3$ anyons would be necessary to cancel braiding effects ($3/5 \times 2\pi \times 5/3 = 2\pi$). Trivial braiding would thus be obtained with anyon pulses containing a charge $q=Ne/5=e/3$ instead of electron pulses in the $\nu=1/3$ case, which is a direct consequence of the non-Laughlin nature of the ground state. Beyond Abelian FQH states, our time-domain measurements can also be applied to the investigation of the dynamical properties of non-Abelian anyons (50, 51).

Coincident with the present investigation, two other works are experimentally addressing the scaling dimension of the $e/3$ FQH quasiparticles at $\nu=1/3$ using a different technique: the thermal to shot-noise crossover of the current fluctuations at the output of a QPC driven by a dc voltage. Schiller *et al.* (52) found $\delta=1$; Veillon *et al.* (53) found $\delta=1/3$, in agreement with expectations at $\nu=1/3$. The differences observed between the various works show the nonuniversality of δ that may be related to differences in the geometry of the QPCs.

Materials and methods

Current noise measurements

The current noise measured at output 4 of the cQPC is converted into a voltage noise through the quantum Hall edge channel resistance $R_{1/3}=3h/e^2$ between the output ohmic contact and the ground. The output ohmic contact is also connected to the ground via a tank circuit with a resonance frequency of 1.1 MHz. By shifting the measurement frequency in the megahertz range, the tank circuit removes parasitic noise contributions that are present at low frequencies. It is followed by a custom-built cryogenic amplifier and a room-temperature amplifier. A vector signal analyzer measures the autocorrelation of the output-voltage noise in a 100-kHz bandwidth centered at 1.1 MHz.

FQH effect and anyon fractional charge

The presence of integer and FQH states is identified by measuring the cQPC's backscattering probability P when the cQPC's gates are energized with a positive gate voltage of 50 mV. A measurement of P as a function of the magnetic field is presented in fig. S3. P goes to zero signaling ballistic propagation at the sample edge for integer quantum Hall states as well as for the fractional state $\nu=1/3$ (where $P=0$ for $B>13.4$ T). Once the magnetic field is set on the $\nu=1/3$ state, P can be tuned by varying the cQPC's gate voltage. In the weak backscattering regime, the anyon charge can be determined from current shot-noise measurements. The current noise measurements for $P<0.3$ presented in fig. S3 are consistent with the expected anyon charge $e/3$ at $\nu=1/3$.

Pulse generation

The step voltage applied to gate G1 is generated at room temperature by an arbitrary wave generator with a sampling frequency of 64 GHz. The voltage pulse propagates toward the sample through a 40-GHz-bandwidth coaxial line with discrete attenuators placed at each stage of the fridge (for a total attenuation of 60 dB). Owing to the dissipation in the coaxial lines, the attenuation increases with frequency, which we compensated for by correcting the step voltage delivered by the arbitrary wave generator.

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Time-domain braiding of anyons

M. Ruelle, E. Frigerio, E. Baudin, J.-M. Berroir, B. Plaçais, B. Grémaud, T. Jonckheere, T. Martin, J. Rech, A. Cavanna, U. Gennser, Y. Jin, G. Ménard, and G. Fève

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Editor's summary

Anyons, particles that acquire a nontrivial phase upon their exchange, can appear in two-dimensional systems such as fractional quantum Hall (FQH) liquids. Ruelle *et al.* studied the dynamics of anyon tunneling by generating single-anyon pulses and sending them toward a constriction in a FQH system. The researchers found that exchange effects slowed down the dynamics of anyon tunneling, in contrast to the situation for electrons. —Jelena Stajic

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